packages/graph-core/src/__tests__/normalize-graph.test.ts

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T05:57:51.904Z

Overview

A new test file `packages/graph-core/src/__tests__/normalize-graph.test.ts` (lines R1-R498) has been added to exercise the graph-normalization utilities and their integration with the schema validator.

Key changes

- **Imports added** (R1-R7):```ts import { describe, it, expect } from "vitest"; import { normalizeNodeId, normalizeComplexity, normalizeBatchOutput, } from "../analyzer/normalize-graph.js"; import { validateGraph } from "../schema.js";

```- **`normalizeNodeId` tests** (R9-R107): cover handling of file, function, and class IDs; stripping of project prefixes; normalization of bare paths; whitespace trimming; support for non-code prefixes; fallback for unknown types.

- **`normalizeComplexity` tests** (R124-R184): verify mapping of string aliases, numeric ranges, case insensitivity, and defaulting to `"moderate"` for undefined, null, zero, or negative values.
- **`normalizeBatchOutput` tests** (R187-R440): assert node ID rewriting, numeric-to-string complexity conversion, edge rewriting, dangling-edge removal, node/edge deduplication, and accurate statistics reporting.
- **Integration test** (R443-R497): builds a graph from the normalized output, validates it with `validateGraph`, and checks that the resulting graph contains the expected nodes and edges.

Impact

- **Correctness**: The tests confirm that normalization logic behaves as specified and that the output satisfies the graph schema.
- **Maintainability**: Centralized edge-case coverage makes future refactors easier to validate.
- **Observability**: Statistics from `normalizeBatchOutput` are exercised, aiding debugging of normalization issues.
- **Compatibility**: The public API of the analyzer remains stable; any breaking change will surface in these tests.

Risks & follow-ups

- **Regression in normalization**: Changes to normalization rules will cause test failures; run `vitest` to verify.
- **Schema drift**: If the schema changes, `validateGraph` may fail; keep the schema and validator in sync.
- **Performance**: Large graphs could expose inefficiencies in ID rewriting or deduplication; monitor test runtimes.
```

- **Missing edge cases**: Current tests cover typical scenarios; consider adding tests for uncommon prefixes or malformed IDs if new features are introduced.